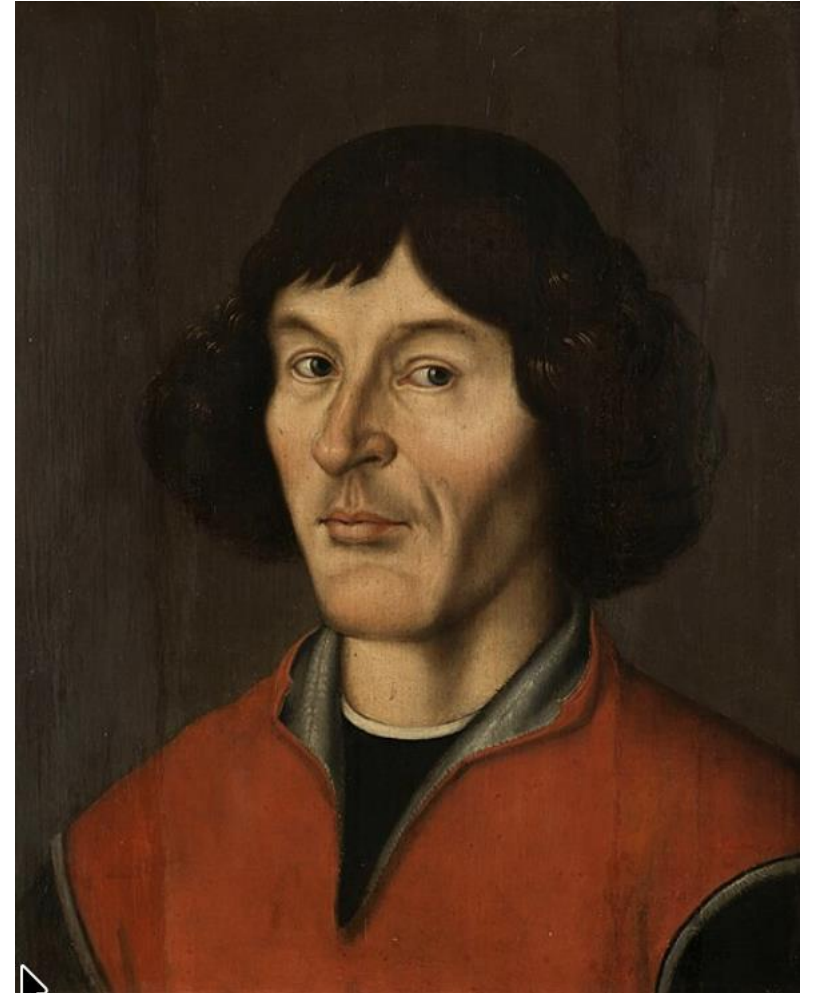


Artificial Intelligence

Eduard Hovy
University of Melbourne

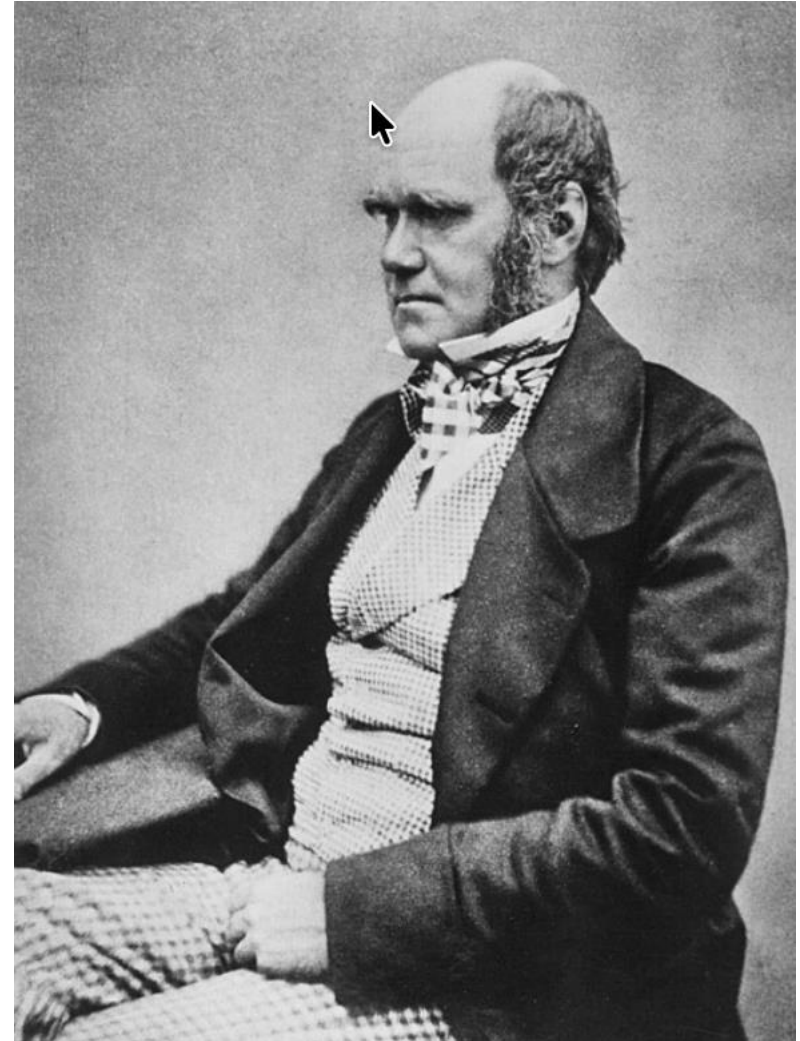
The cosmological insult

- The human is at the centre of the cosmos
- Heliocentric model of the solar system
- Copernicus



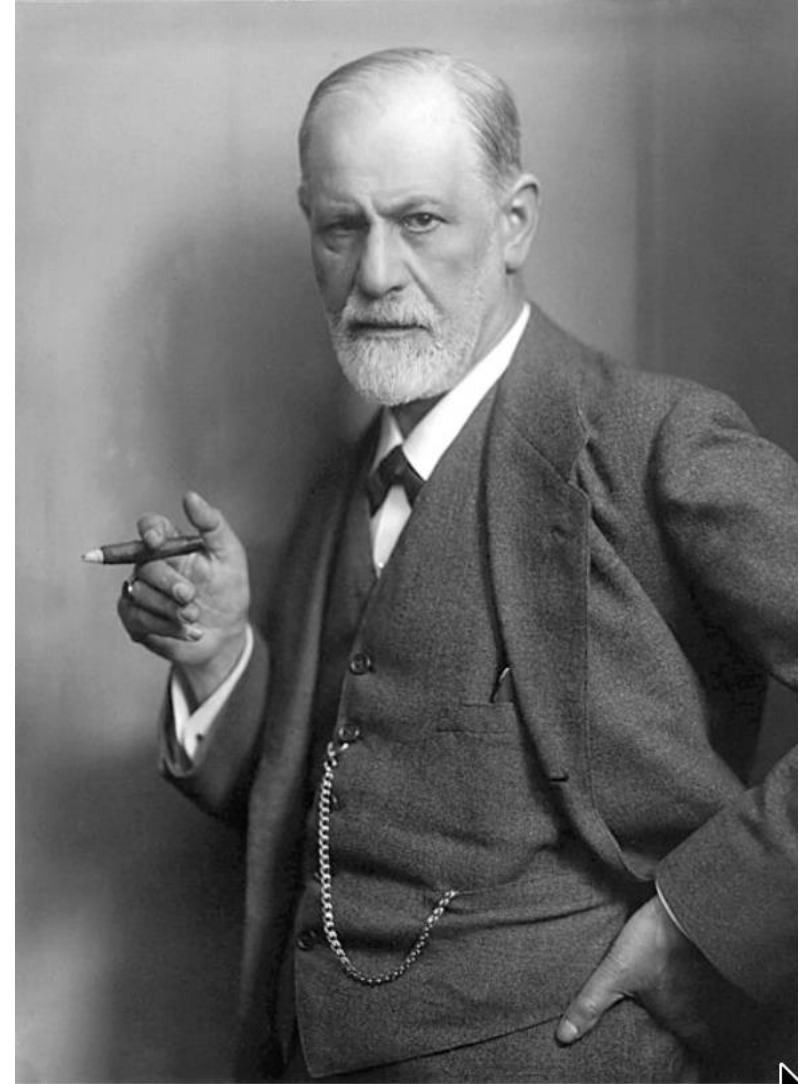
The biological insult

- The human is the pinnacle of life
- Evolution
- Charles Darwin



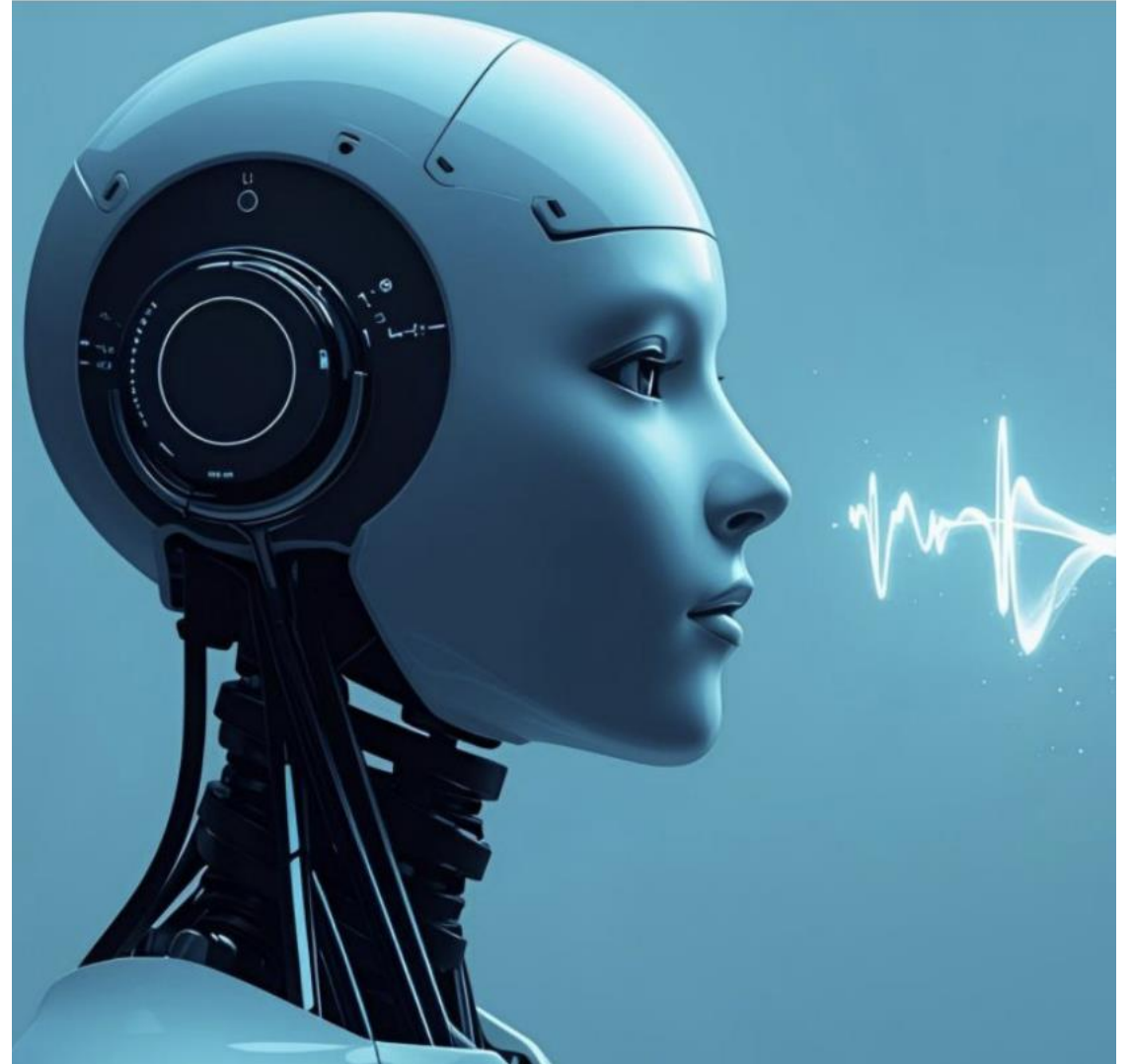
The psychological insult

- The human mind is the apex of thought
- Unconscious thinking
- Sigmund Freud



The intellectual insult

- Only humans can reason
(and be conscious?)
- Artificial Intelligence



Overview

1. What is AI?
2. Some important ideas in AI
3. Neural networks and LLMs
4. Where is AI going?

What is AI?

Is the sphex wasp intelligent?



Is your dog intelligent?

The Turing test — Alan Turing



Eliza — the first chatbot, 1967

<https://psych.fullerton.edu/mbirnbaum/psych101/eliza.htm>

Eliza, the Rogerian Therapist

ELIZA is a computer program that emulates a Rogerian psychotherapist. Just type your questions and concerns and hit return. Eliza will answer you.

Talk to Eliza

```
* hello
> How are you today.. What would you like to discuss?
* I am interested in AI
> How long have you been interested in AI?
* Since birth
> Tell me more...
* I have been interested in AI since birth
> I see.
* So do I
> OK... "I are interested in AI". Tell me more.
```

Input:

When the original ELIZA first appeared in the 60's, some people actually mistook her for human. The illusion of intelligence works best, however, if you li your life.

This javascript version of ELIZA was originally written by [Michal Wallace](#) and significantly enhanced by [George Dunlop](#).

How Eliza works

Is Eliza intelligent?

```
SAD-WORD := {sad unhappy angry miserable...}
```

```
HAPPY-WORD := {happy excited great...}
```

```
LOOP: GET (INPUT)
```

```
IF
```

```
  INPUT includes {SAD-WORD} THEN OUTPUT := "I am sorry to hear you are sad. Why?"
```

```
  INPUT includes {HAPPY-WORD} THEN OUTPUT := "I am glad to hear you are happy. Tell me more"
```

```
  INPUT includes "I am ?X" THEN OUTPUT := "How long have you been ?X ?"
```

```
ELSE OUTPUT := "Go on..."
```

```
WRITE (OUTPUT)
```

```
GOTO LOOP
```

History

- Prehistory
- **1956**, birth of AI: The Dartmouth workshop
- **1960–70, the Wild West of AI**: Augmented Transition Networks, Expert Systems, Speech Recognition and Information Retrieval
- **1980–90, the time of specialization**: Natural Language Processing, early Machine Learning, Knowledge Representation, Robotics...
- **1990–2010, the Statistics Wars**: Machine Learning wins
- **2010–, Neural Networks**: Generative AI, Large Language Models

Areas of AI

- | | |
|--------------|---|
| Think | • KR&R: Knowledge Representation and Reasoning |
| Learn | • ML: Machine Learning |
| Communicate | • NLP: Natural Language Processing (Machine Translation, Info Retrieval, Speech Recognition, Summarisation, QA, Dialogue...) |
| Move | • Robotics (hardware + software) |
| Plan/play | • Games , optimization, search, planning |
| Communicate | • HCI: Human-Computer Interfaces |
| Act together | • Agents |
| Other | • Other (Ethics and Social Impact, Artificial Life algorithms, etc.) |
| | • Applications (Biomedicine, Government, Defence, Engineering...) |

Part 2: Some important ideas in AI

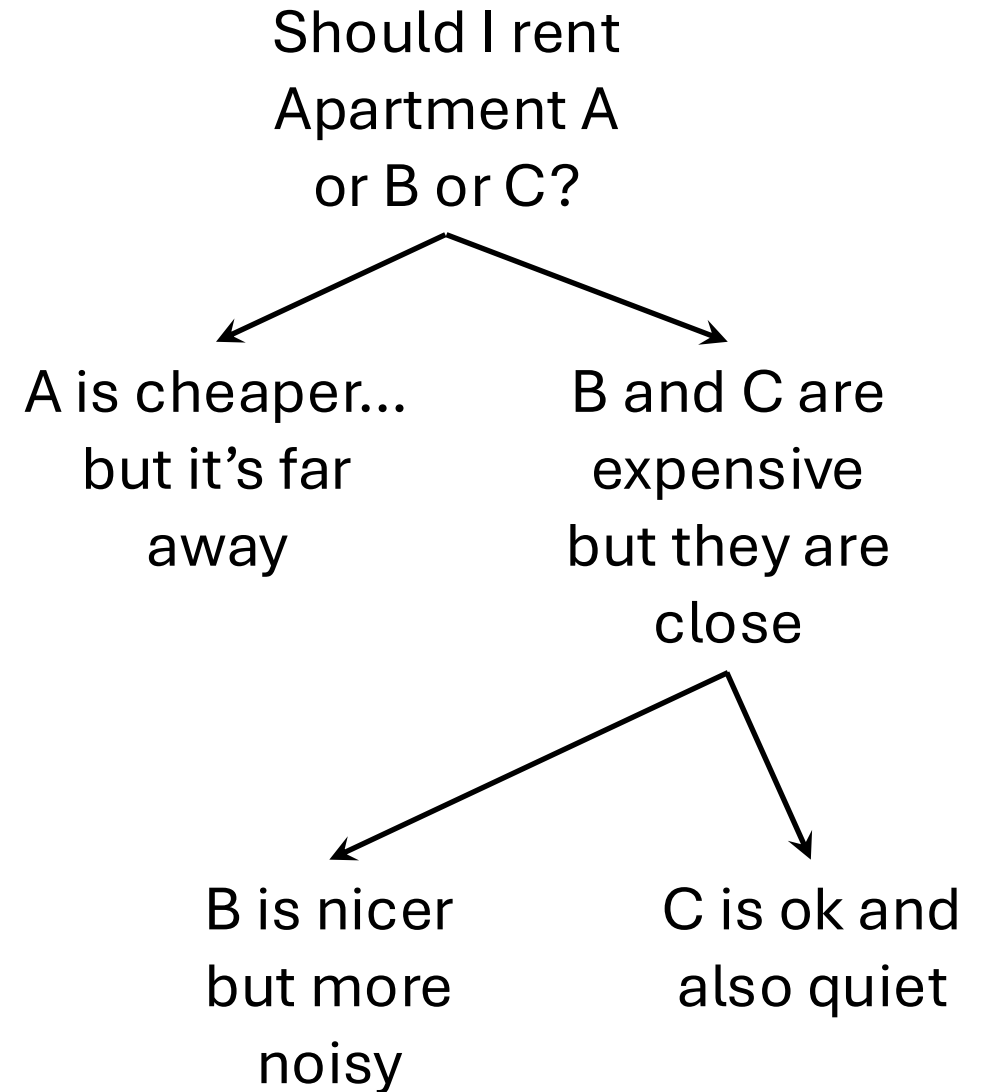
3 Core Concepts

1. Decision Trees and Expert Systems
2. Vector spaces and Information Retrieval
3. Machine Learning

Concept 1: Decision Trees

When you have a complex decision,
you consider the most important aspect
and depending on that,
you consider the second-most ...
...until you reach the small points
and then you make your decision

This is a **Decision Tree**



Expert systems used decision trees

- Very successful in early AI
 - Medical diagnosis
 - identifying unknown organic molecules
 - Chess
 - Fault detection
 - Financial forecasting
- BUT ... the world is too complex to model by hand!
- Decision Trees can be learned from data!
 - Model the relevant factors and the decision
 - Expert systems have been subsumed by machine learning

Concept 2: Vector spaces

- Situation: You have tons of data, often unstructured
 - Each data is a collection of pieces of information
 - Examples: documents of words, images of rooms or outside, tax records for a company
- Problem: How can you organize the data and find things — especially when the data is not nicely ordered?
- Case example: the internet. How does Google work?



Encoding documents as vectors (word lists)

- You can represent a document as a vector of words:

	change	school	degree	year	future	mind	eat	cheese	oil	...	words
doc001	2	4	1	4	3	1	0	0	0	...	
doc002	1	3	3	1	4	2	1	0	1	...	documents
doc003	0	2	0	0	2	3	3	4	2	...	

STUDIES **FOOD**

- Typical vector length ~100,000 – 250,000 words/symbols
- The numbers:
 - word frequencies (= simple word count)
 - Or *tf.idf* scores (penalize frequent words that appear everywhere, like “the”)

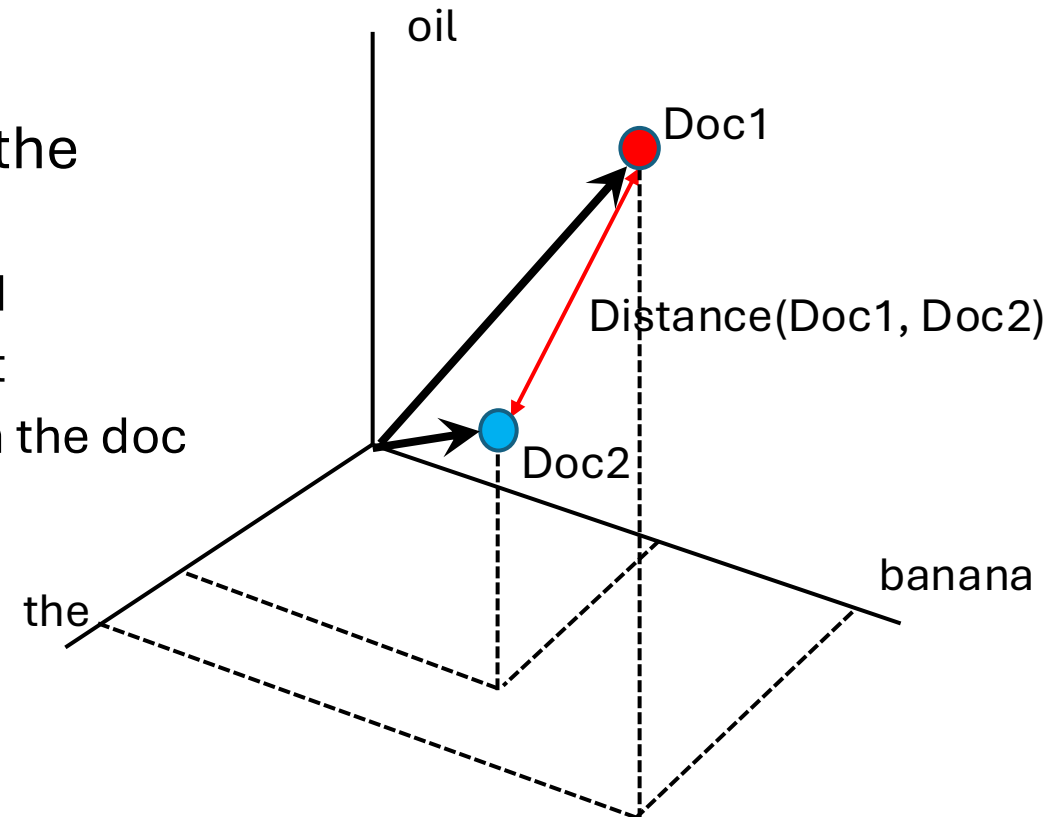
$$tf(t, d) = \frac{f_{t,d}}{\sum_{t' \in d} f_{t',d}},$$
$$idf(t, D) = \log \frac{IV}{|\{d : d \in D \text{ and } t \in d\}|}$$

Vector spaces

Goal: Automatically find relevant texts and cluster similar ones

Approach: Build a large 'vector space' to store the documents

- Each axis ('dimension') stands for a different word
- Each point in the space represents one document
- Point's position is given by a vector of the words in the doc
- You can measure the distance between docs
- Similar-meaning docs are close together



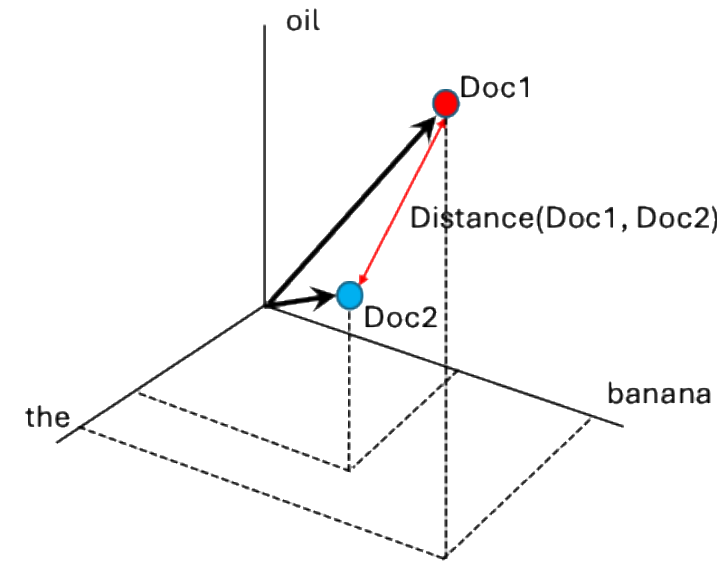
How Google works

Building (what Google does):

1. Collect billions of documents from the web (= “crawling”)
2. Extract the words and counts them (= “scraping”)
3. Place each document in the Vector Space (= “indexing”)
4. Upweight popular documents (= PageRank algorithm)

Usage (what you do):

1. Submit your query
2. Query words indicate a region in the Vector Space
3. The documents in that region are sorted and returned to you



Concept 3: Machine Learning

Learning plumbing



Is the tap leaking? If so,
get a new one

Is the pipe cracked? If
so, replace it

Is the gasket old? If so,
take it out

...

- Many seemingly ‘intelligent’ tasks can be broken down into a sequence of simple steps
- Each step requires a decision based on the situation
- The situation —and its decisions— can be characterized by a set of pertinent features
- People make decision based on the features they observe
- The decision ‘rules’ can be learned automatically by observing a large number of examples
- Machine learning algorithms look for patterns in the feature that predict the actions — the ‘rules’

Machine learning methodology

You collect a large number of examples: situations + desired decision

You define the features (F1, F2...) to be considered, and for each one you build a feature recognizer (small algorithm or sensor)

Data prep

	F1	F2	F3	F4	F5	F6	F7	DECISION
Ex 1								
Ex 2								

You build a statistical algorithm that integrates all examples to find the typical decision for each combination of features

Training

You take a new collection of examples, not yet seen

You test the algorithm to see how well it does

Evaluation

Example: Who should you lend money to?

Training

<i>Tid</i>	Refund	Marital Status	Taxable Income	Cheat
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

Testing

Tid	Home Owner	Marital status	Annual Income	Defaulted Borrower
11	No	Single	55K	?

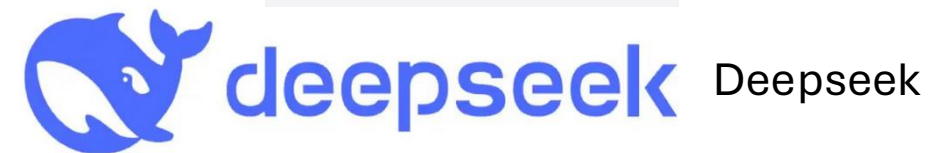
Machine learning today

- Dozens of families of ML algorithms
 - Many available on the web
- Thousands of datasets on the web and in companies, hundreds of domains
 - Weather, agriculture, engineering, medicine, many research areas...
- Several outstanding successes
 - New chemicals, protein folding (AlphaFold), COVID outbreak prediction...
 - Fraud detection, credit card problems, credit risk assessment...
 - Recommender systems for music, products, travel...
 - Self-driving cars, robots that learn, supply chain optimization, workflow optimization...
- Latest development: Neural Network learning

Part 3: Neural networks and LLMs

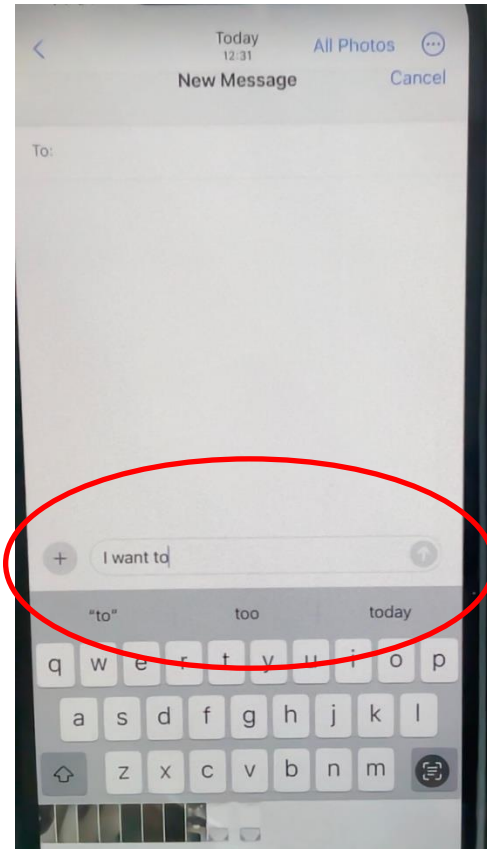
Large Language Models (LLMs)

- What are they?
- What can they do and not do?



Meta

Language Models



Your cellphone can predict what you are about to type

How does it know this?

It needs a list of possible (different) continuations

Context sensitivity
using lists of words:
“... to <?>”



w1	w2	w3	score
want	to	go	0.000391
want	to	eat	0.000015
...			
go	to	the	0.000491

Building a neural network LM

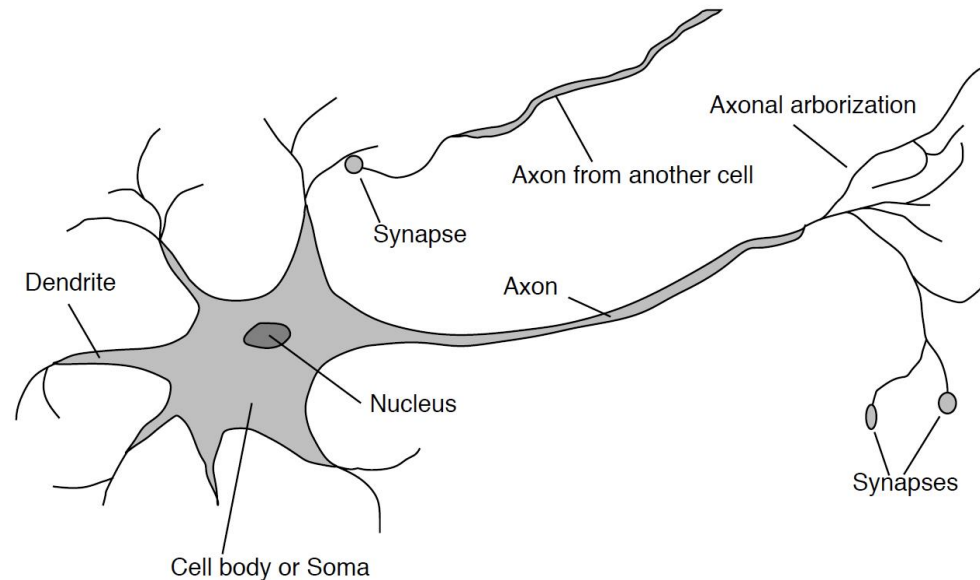
A LM is a list of word sequences that you learn from data

A neural network is a very powerful machine learning system

How to learn the words sequences in a neural network?

What is a neural network?

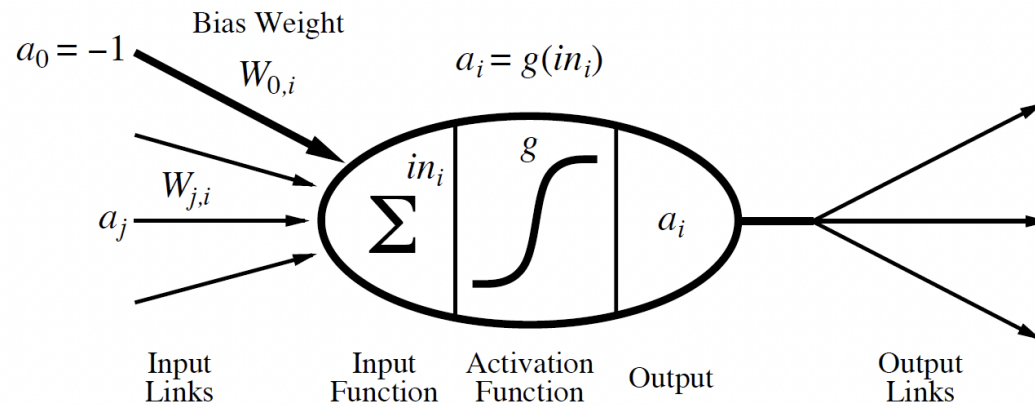
- Human brain contains ~86 billion neurons connected by axons in a vast network



- Each neuron performs a very small ‘computation’ based on its inputs to produce outputs that activate other neurons — ‘activation flow’
- The strengths of activations are adapted by learning

AI approximation

Define a ‘neuron’ as a small computational function that combines its input values (a vector of real numbers) to produce an output value (also a vector of real numbers)



- What function?
- What connections (= network structure)?
- How to adjust/train the weightings (= strengths) of the connections?

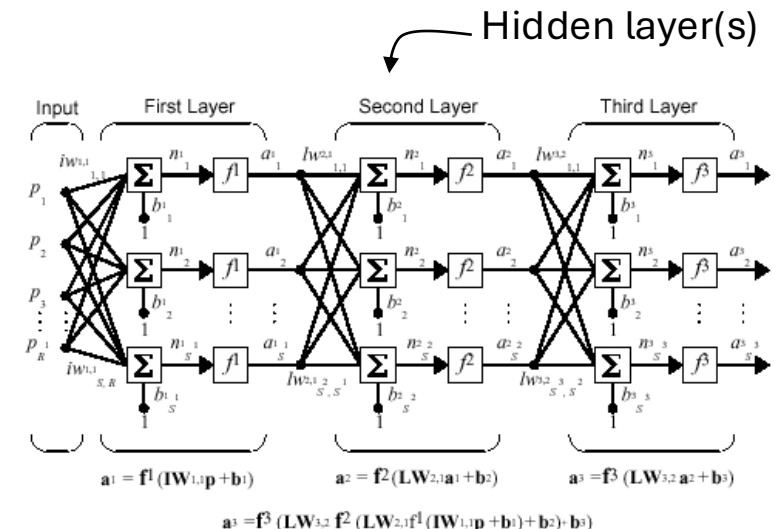
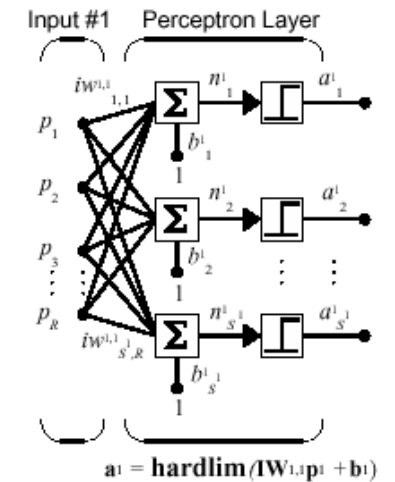
Some network structures

A NN contains dozens to hundreds of nodes

They are arranged in various ways (layered, perhaps with loops, etc.)

Each arrangement gives different behaviours

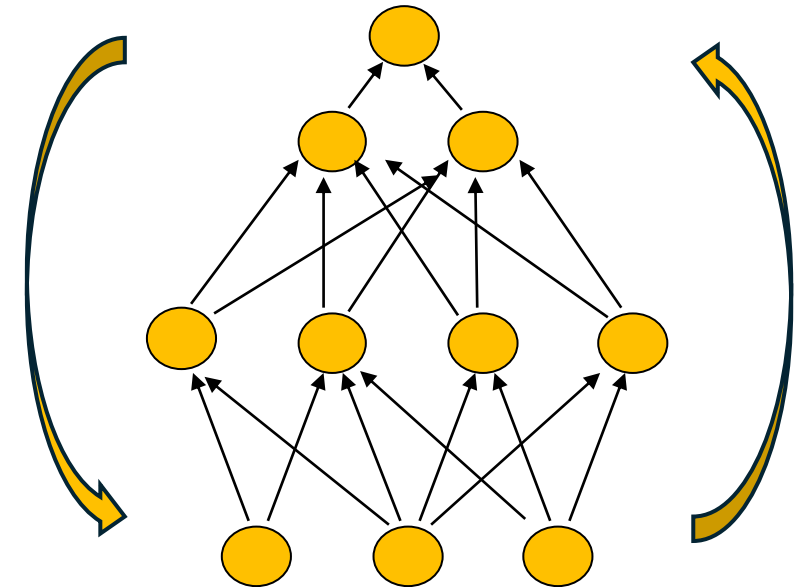
- Perceptron
 - The simplest one-node example
- Feedforward
 - Signal goes through in one direction, no loops
- Recursive and Recurrent
 - Signal can loop back and reinforce itself
- Convolutional
 - Small local input groups can be treated together
- Transformer
 - The most popular one today (like ChatGPT and Claude)



Training and using neural networks

Construction:

1. Encode the data as numerical vectors
2. Build a neural network:
 - Choose number of nodes in each layer
 - Choose number of layers
 - Create connections between nodes (= links)
 - Choose the numerical combination function for the nodes
 - Define how to interpret and use the final output node(s)
3. Train the neural network using many input-output examples to adjust the node function's parameters (= weights on the links)



Use: Feed it new vectors to calculate results for you

Training a NN example: The iris

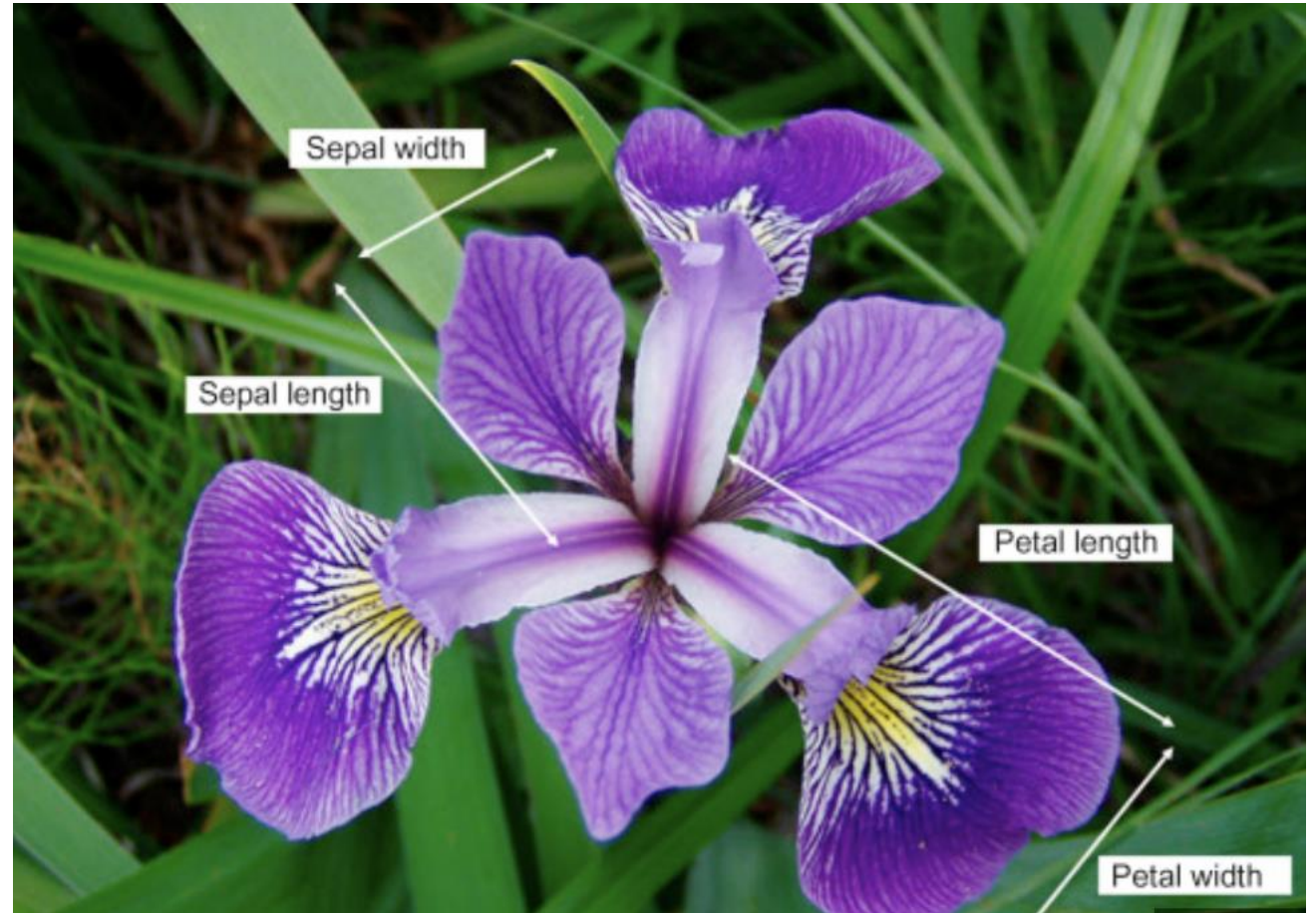
Say we want to classify irises

Three types:

- Setosa
- Virginica
- Versicolour

Four features:

- Petal length
- Petal width
- Sepal length
- Sepal width



Well-known dataset introduced by statistician Ronald Fisher with 150 objects (https://en.wikipedia.org/wiki/Iris_flower_data_set)

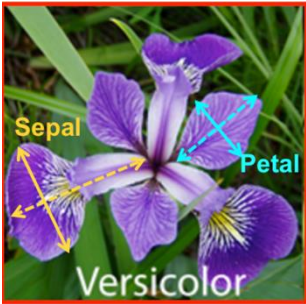
Our data

Sepal length	Sepal width	Petal length	Petal width	Species
5.1	3.5	1.4	0.2	setosa
4.9	3.0	1.4	0.2	setosa
4.7	3.2	1.3	0.2	setosa
4.6	3.1	1.5	0.2	setosa
5.0	3.6	1.4	0.2	setosa
5.7	2.9	4.2	1.3	versicolour
6.2	2.9	4.3	1.3	versicolour
5.1	2.5	3.0	1.1	versicolour
5.7	2.8	4.1	1.3	versicolour
7.6	3.0	6.6	2.1	virginica
4.9	2.5	4.5	1.7	virginica
7.3	2.9	6.3	1.8	virginica
6.7	2.5	5.8	1.8	virginica

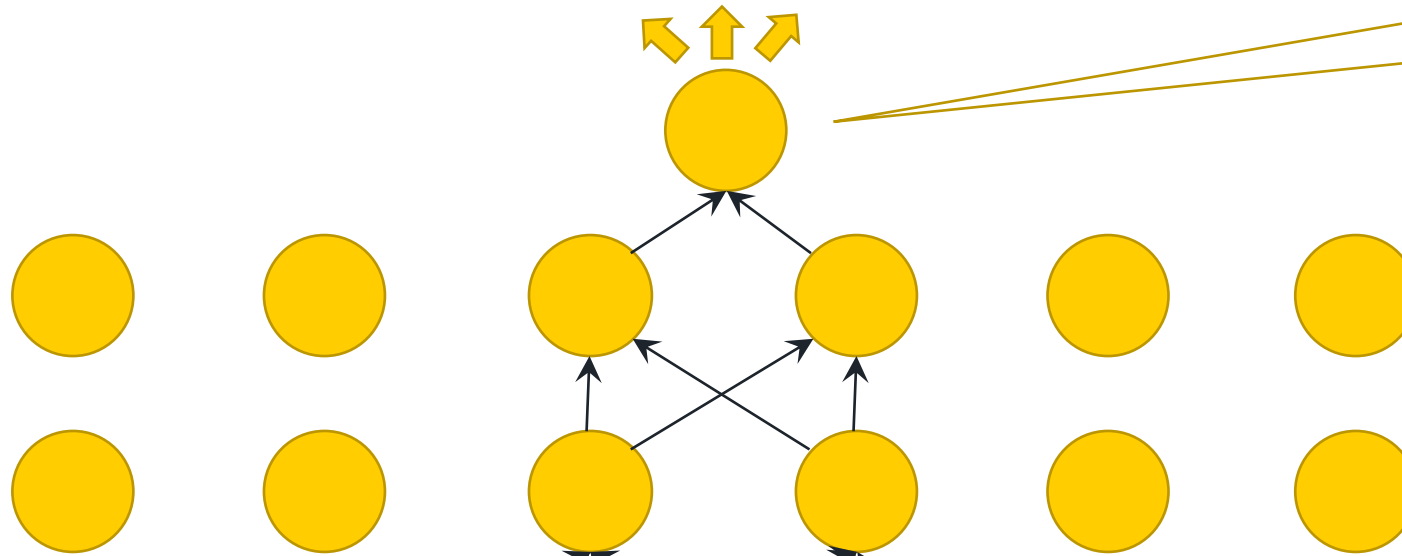
Setosa: Short, narrow petals,
but sepal info not distinguishing

Versicolour: Petals kind-of in the middle
but sepal info not distinguishing

Virginica: Longer petals,
but sepal info not distinguishing



Learning the patterns



If $R > A$ then *setosa*
else if $R < B$ then
versicolor else *virginica*

Actually, ~12 layers,
several dozen to 100
nodes in a row

$$R = a_1SL + a_2SW + a_3PL + a_4PW$$

Actually, a nonlinear
function like *tanh*

Sepal length SL	Sepal width SW	Petal length PL	Petal width PW	Species
5.4	3.1	2.1	0.5	setosa

Tens of thousands of
examples, for training

Moving to words: Large Language Models

L="station" → train-related

L="the" → noun

L="to the" → location

L="on" → location



"platform"

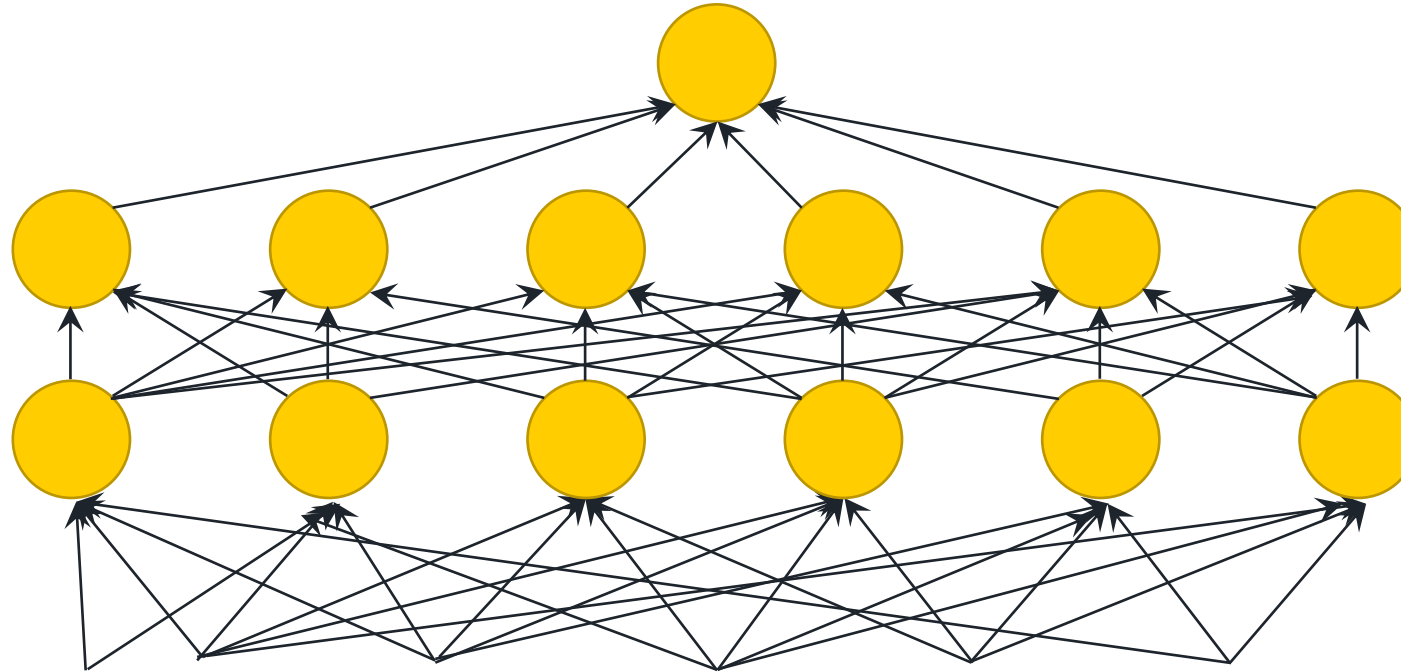


R="train" → train-related

R="get on" → near vehicle

R="9¾" → "Platform"

R="wait for" → location



"at the station he went upstairs to the _____ and waited for the train"

"At King's Cross Station on _____ 9¾ you can get the Hogwarts Express"

Abstract 'microfeatures'

Weight scores on each
link adjusted during
learning:
"backpropagation"

Hundreds of millions of
sentences
Each word is a vector

Producing sentences: Generative LLMs

What is the future of LLMs in government?

Future governments will...

In future, government offices ...

Government workers need to...

...

...

government offices are likely to start ...

government offices will have to ensure ...

government offices are certainly going to ...

...

...

are likely to start using LLMs

using LLMs in their daily

in their daily work at ...

...and just keep
going, left to right

So what do we have?

A very big database of (implicit) text fragments with a chat loop

No understanding of truth:

- May happily report something it has read that is false (flat earth problem)
- May assemble parts of two sentences that are separately true into something that is not

Cannot reason:

- It cannot infer, check consistency, or anything that requires logic
- What apparent logic there is comes from the long context (the matching-window)

No goals or wishes:

- It cannot 'want' anything, like to take over the world
- You can easily 'drive' it to astrology, hatred, racism, or anything else it has read about

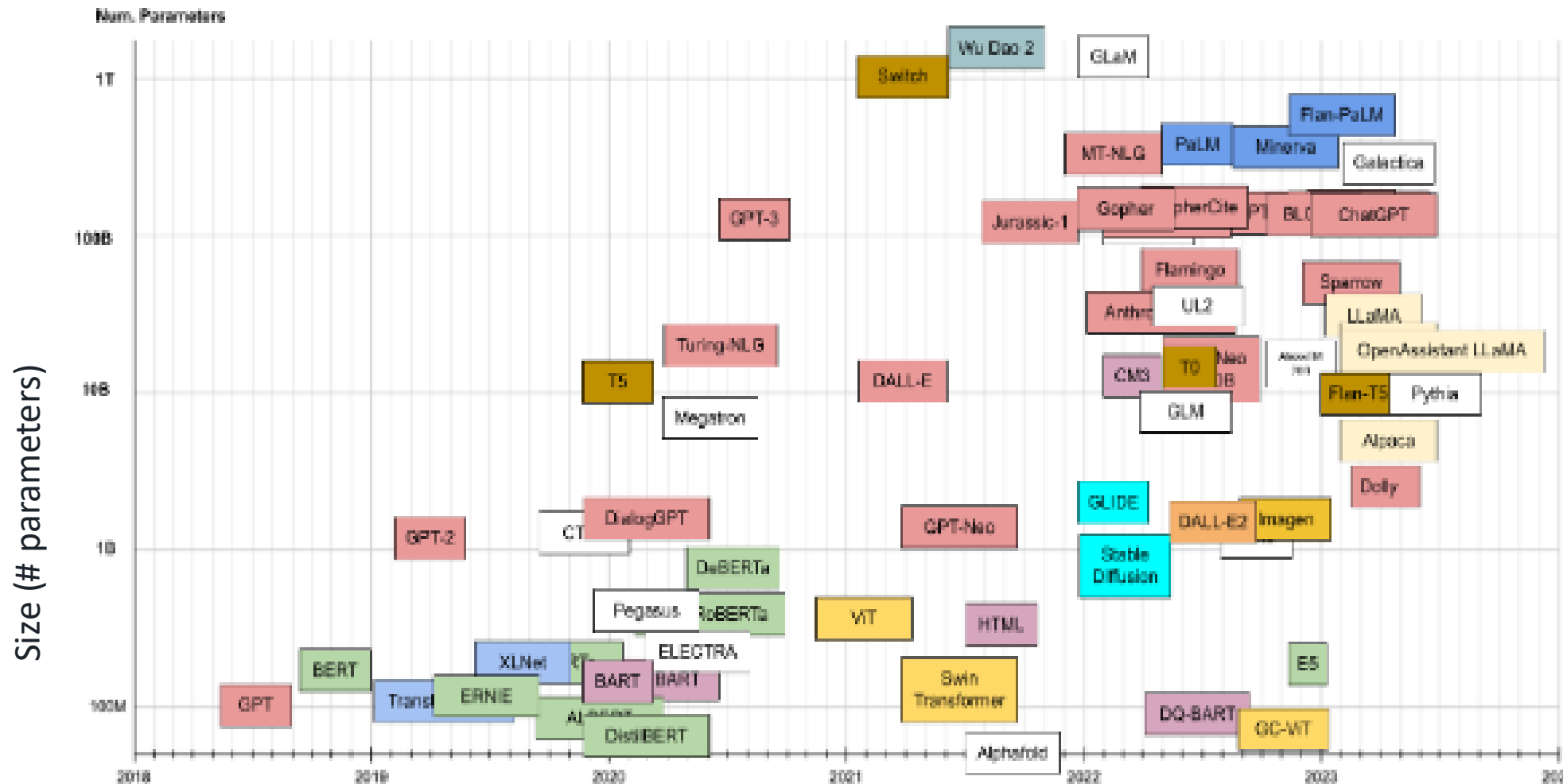


This machine has no brain

Use your own

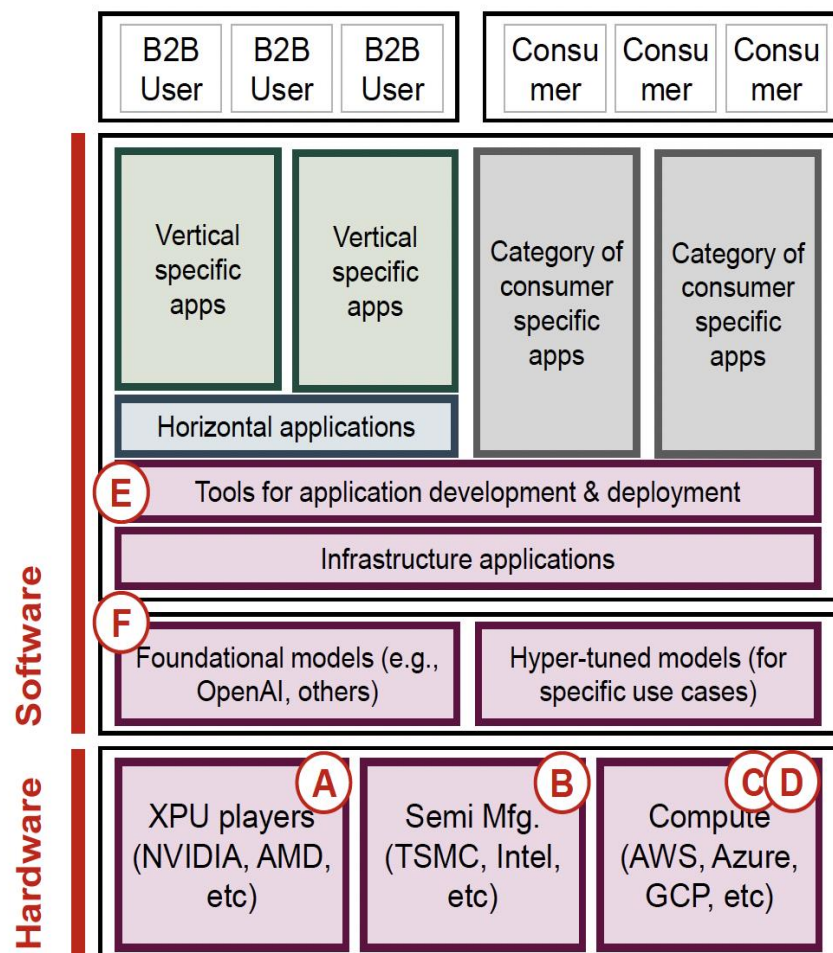
Part 4: Where is AI going?

More and more LLMs



Colours describe type of DNN Transformer

More and more layers of functionality



Hardware and infrastructure

(A) Specialized compute and interconnect: NVIDIA, BROADCOM, MARVELL

(B) Leading edge semiconductor manufacturing: TSMC

(C) Large HPC cluster/ hosted service players:



(D) Edge compute/ retraining/ inference players: many nascent players

(E) Machine learning and development tools

Data prep: Potential need for different data tools (e.g., vector database tuned for embeddings, a different approach to labeling) but also more need for existing data tools

Model development:

- ML frameworks for efficient training:
- Model experiment mgmt., & collaboration:



Production: Potential need for different types of Model deployment, observability and backend/orchestration to existing MLOps tooling



(F) Foundational models



Cost of usage

March 2023: GPT-4 cost ~US\$50 to work with a million tokens

August 2025: GPT-5 **cost ~14c per million tokens**

- (And GPT-5 is much more capable than GPT-4)

2025: Marginal **energy** for a standard prompt of a modern LLM is approx. 0.0003 kWh

- Same energy use as 8–10 seconds of streaming Netflix
- Equivalent of a **Google search in 2008**

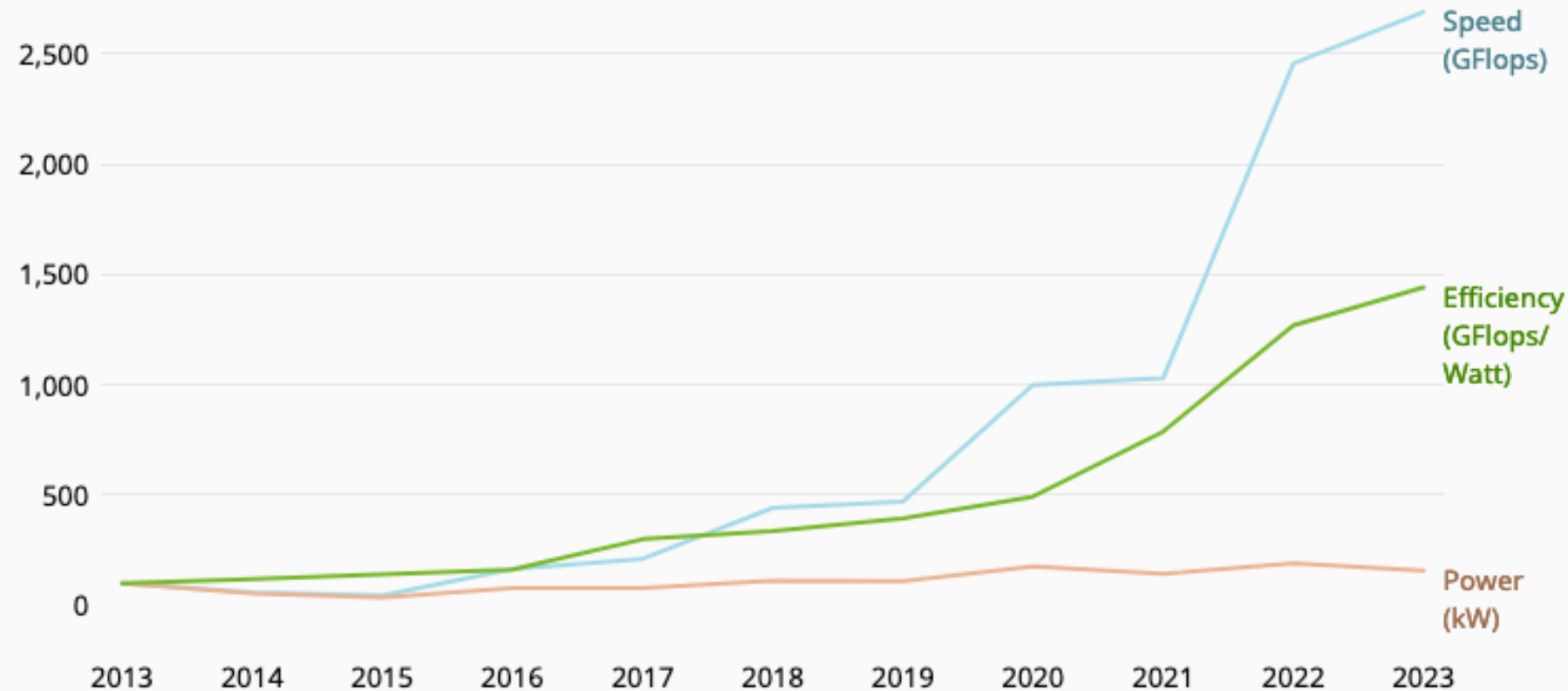
2025: **Water** used ranges from a few drops to **1/5 of a shot glass** (.25mL to 5mL+)



https://x.com/emollick/status/1959989512228208785?utm_source=substack&utm_medium=email
both independent tests and official announcements

Energy and efficiency

Figure 8 Average speed, power and efficiency of top 50 green supercomputers, 2013–2023



So the AI future is bright

You can buy Microsoft Copilot or similar

You get a big package with lots of functionality, and periodic updates

You can run things yourself, with your own RAG store

But it costs a lot

You can contract with OpenAI, Anthropic, etc. or with AWS, Salesforce, etc.

You can assemble your own selection of GenLLMs and run them privately

You can run things on you own Cloud space, with your own RAG store

But it costs a lot

You can contract with smaller vendors, or hire your own specialists

You can download free GenLLMs and/or build and train your own

You have complete on-premises capability for total privacy and safety

This will cost less over time, but is probably not feasible for most people for another few years

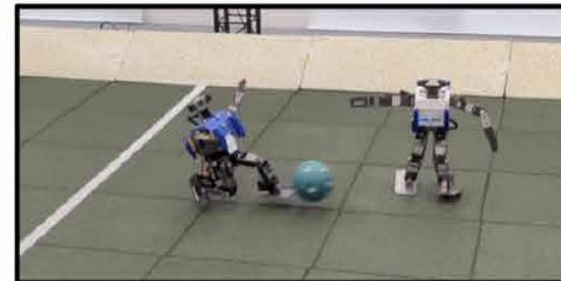
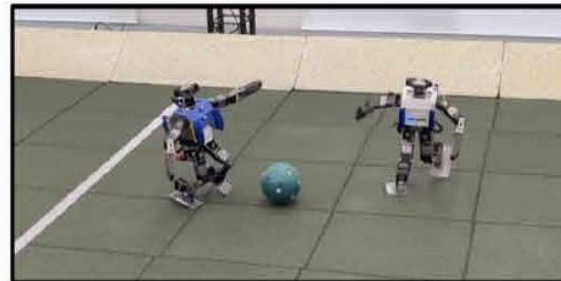


Renting the
equivalent costs
\$20k/month today

H100 machine + lots
of memory costs
\$250k today

Robots

- Bipedal robot movement has been a great challenge for many years
- Many degrees of freedom of movement (direction, turning, upright balance, etc.)
- Soccer is the acid test
- Study used deep reinforcement learning (deep RL) to synthesize sophisticated and safe movement skills for a low-cost, miniature humanoid robot
- Result is robust and dynamic movement skills: rapid fall recovery, walking, turning, kicking, and smooth transitions. Also learned to anticipate ball movements and block opponent shots
- Trained with a combination of high-frequency control, targeted dynamics randomization, and perturbations during training (like knocking the robot over by hand)
- Results: Faster walking (181%), turning (302%), getting up (63%), kicking ball (34%) than a script



AI capabilities: Automated drivers

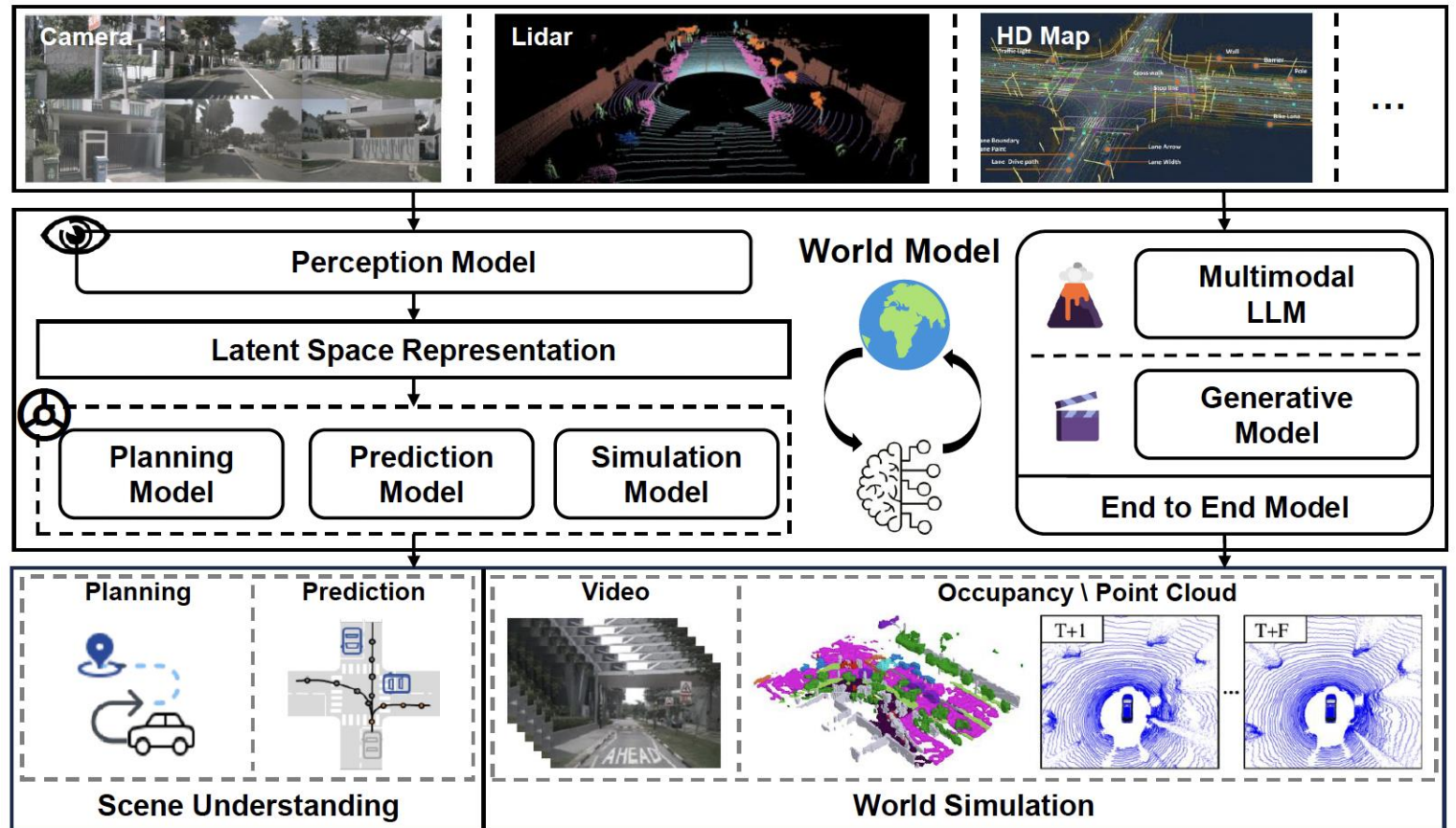
- Waymo automated driving is 90% better than people
 - 4 cities in the US
 - Accidents with *any injury reported* rate:
 - Automated Driving Systems: 0.6 incidents per million miles (IPMM)
 - Human drivers: 2.80 IPMM
 - 80% reduction or a human crash rate that is 5 times higher than the ADS rate
 - ~50% of automated driver accidents were at <1 mph
- Drivers in Australia
 - About 70,000 taxi drivers
 - About 180,000 truck drivers
- What work will these people do?
 - K. Kusano et al. 2024. Comparison of Waymo rider-only crash data to human benchmarks at 7.1 million miles. National Library of Medicine. See <https://pubmed.ncbi.nlm.nih.gov/39485678/>
 - Wikipedia
 - Australian Government Jobs and Skills report <https://www.jobsandskills.gov.au/data/occupation-and-industry-profiles/occupations/7331-truck-drivers>

World models and Foundation Models

Adding more and more knowledge...

Text + images + speech + video + robot body sensing + emotions + brain signals...

Example for
autonomous driving:
(Ding et al. 2025)



Emotion and empathy

ChatGPT and other models are better at empathy and emotion than you'd think!

EmotionPrompt: Adding emotion-type info consistently improves GenLLM performance

- 4 models (incl. ChatGPT)
- 9 tasks: Truthfulness, Sentiment Analysis (SA), Sentence Similarity (SS), Cause Selection (CS), Sum of numbers (Sum), Word in Context meaning (WC), First Letter (FL), etc.
- Emotion: Use phrases that strengthen (1) self-esteem or (2) social identity
- Examples: “are you sure” “this is very important to my career” “stay focused” “take pride in your work”

Li et al. 2023.

EmotionPrompt: arXiv

ChatGPT-4 vs **Advice Columnists:** People prefer the machine!

- 50 social dilemma questions randomly selected from 10 well-known advice columns (*Dear Abby* etc.)
- Subjects shown 2 questions + 2 responses: one columnist and 1 ChatGPT-4
- Subjects could identify ChatGPT responses, but still preferred them over columnist responses (56%)

Howe et al. 2023.

Frontiers in Psychology
(Cognitive Science)

Three major directions for New NLP

NLP engineering: Make LLMs usable

- Build smaller and cheaper LLMs
- Systematize prompt engineering
- Integrate language, images, and other media and functions

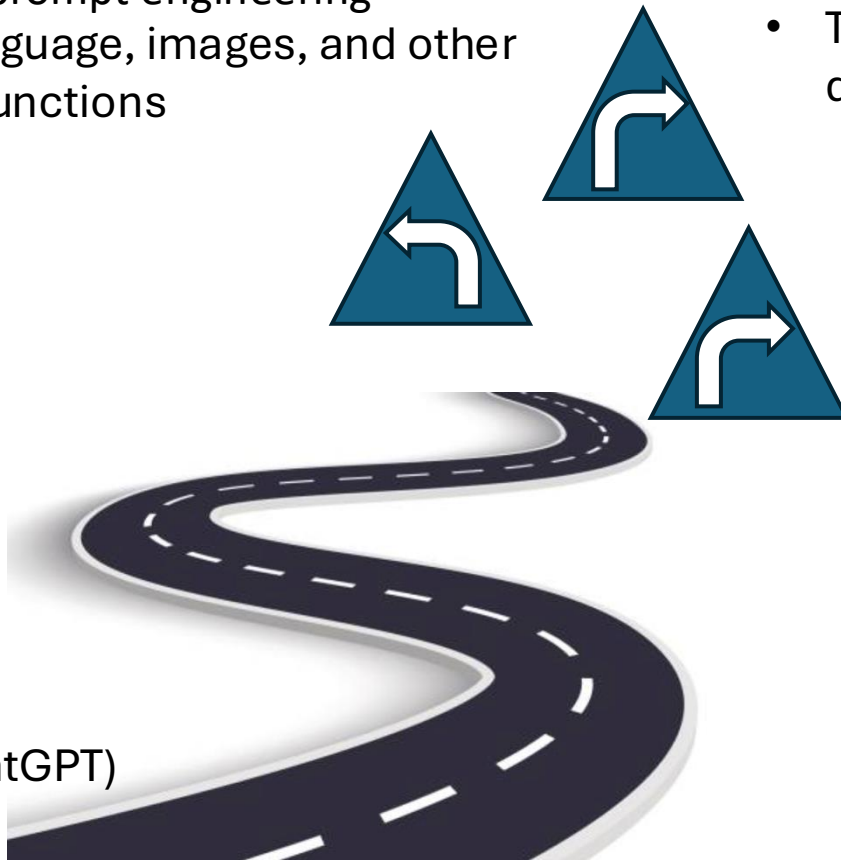
NLP applications: Make LLMs useful

- Tune LLMs to domains and companies for enterprise processing
- Add functionality and agency in the world
- Tailor LLMs to people to be their personal daemons/amanuenses in everyday life

NLP research: Make LLMs understandable

(or at least, be solid engineering)

- Fix the problems with LLMs
- Get explanations how LLMs do what they do
- Formalize them well enough for autonomy, assurance, and ethics



BC (Before ChatGPT)

Conclusion

GenAI is being used, but is not ‘taking over’

- For some tasks it is very helpful
- For some tasks it is useless
- For many tasks the jury is still out

As of November 2025, 12% of U.S. employees use AI daily in their role

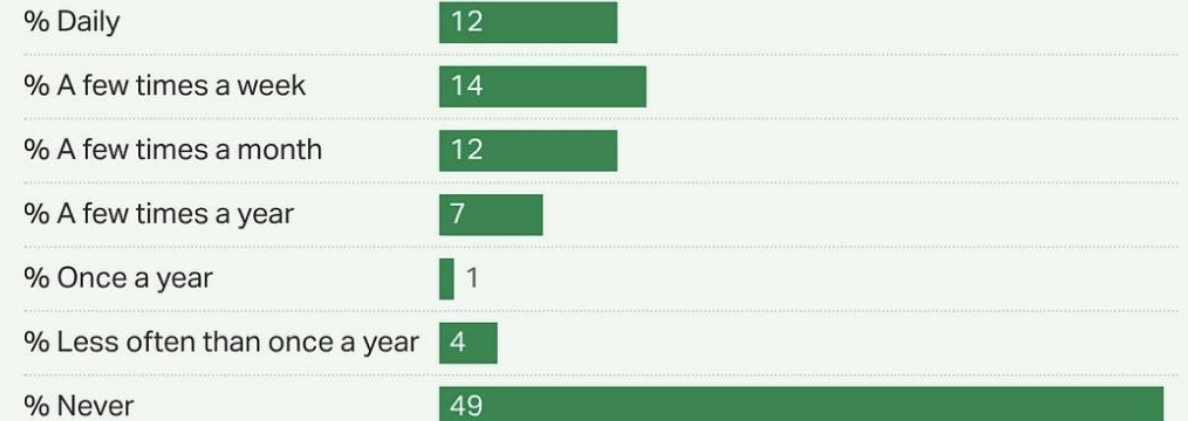
Snapshot of Q4 2025

How often do you use artificial intelligence in your role?

Use the tabs below to view AI use as grouped trends over time or by response frequency.

U.S. Trend

Q4 2025 Snapshot



Social disruption, ethics, and government control

- Certainly AI is being used, and will be used, to ‘bad’ ends
 - (Errors and hallucination)
 - Cyber-hacking
 - Disinformation
 - Job loss
- Government regulation: the US, EU, and Australia
- What are our options, realistically?

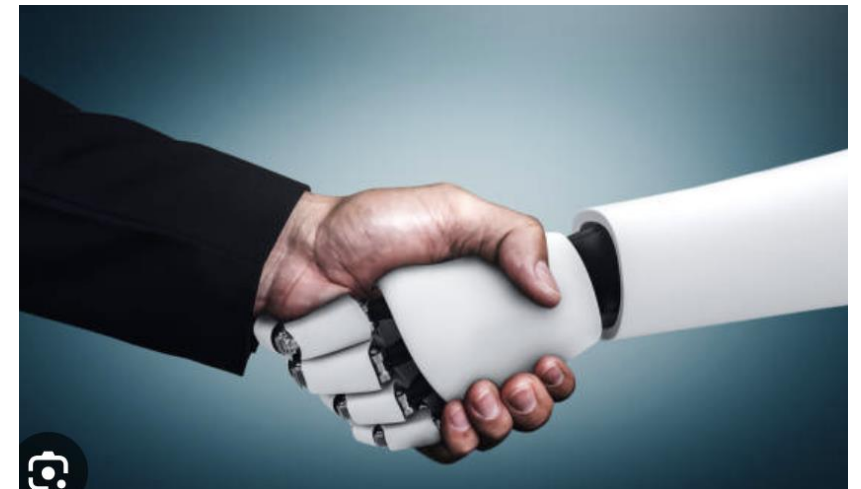
Is AI going to evolve into AGI (General Intelligence)?

Hahaha ... Please define AGI!

- Might AI even kill people? Probably yes ... every technology does. But there has never been a technology with so much publicity and oversight so early, not even nuclear
- Is AI going to be better than humans? Sure, for many things ... many apps and tech already are. But that doesn't replace people
- Is AI going to control society? How? What are its levers of power? *People* control society, not AI. And they are already using AI

Summary: Long-term trends

- AI will boost productivity across the board, some places more than others
- But that is quickly copied and commoditized so it won't create a lasting effect
- Tech stack convergence (open models, shared data/algorithms, cheaper hardware, abundant talent) and limited defensibility of proprietary data will drive homogenization
- Durable advantage: the “residual heterogeneity” (= human factors): human creativity, relationships, novel business design, distinctive distribution/partnerships, and original customer experiences built *around* (not just with) AI



We tend to overestimate the effect of technology in the short run and underestimate the effect in the long run — Roy Amara

Thank you